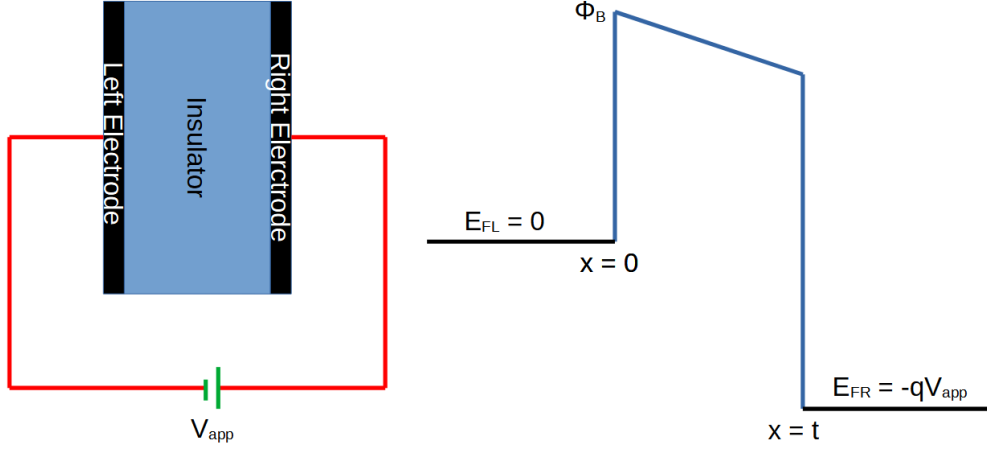


# Hackathon 2

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Consider an RRAM structure whose device and band diagram schematics are shown below.



The direct tunnelling current density from left to right is given by

$$I = \frac{4\pi m_{DOS}^* q k_B T}{h^3} \int_{E_{FL}}^{\phi_B} T(E) \ln(1 + \exp\left(-\frac{E - E_{FL}}{k_B T}\right)) dE$$

$$T(E) = \exp\left(-\frac{2}{\hbar} \sqrt{m_{tun}^*} \int_0^t (E_C(x) - E) dx\right)$$

| Parameter   | Value                        | Units |
|-------------|------------------------------|-------|
| 1           | $1.6 \times 10^{-19}$        | C     |
| $m_{DOS}^*$ | 1                            | $m_0$ |
| $m_{Tun}^*$ | 0.5                          | $m_0$ |
| $m_0$       | $9.1 \times 10^{-31}$        | kg    |
| $T$         | 300.0                        | K     |
| $h$         | $6.62607015 \times 10^{-34}$ | J-sec |
| $\hbar$     | $1.05457 \times 10^{-34}$    | J-sec |

Please calculate the current from the left-to-right electrode for the given  $E_C$  v/s x profile